

CORNELIUS MULYOKELA

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EDUCATION

Stetson University

B.S. Computer Science & Applied Mathematics (Data Science)

GPA: 3.83/4.0

Honors: Pi Mu Epsilon — Inducted for excellence in mathematics, 2025

DeLand, FL

Expected Aug 2027

SKILLS

Languages: Python, Java, C#, C++, JavaScript, HTML, CSS, SQL, MATLAB, R

Technologies: Kafka, Spark, Airflow, PostgreSQL, ETL & data pipelines, AWS (S3, EC2), Flask, Tableau, MySQL, NumPy, Docker, Linux, Bootstrap, React, Node.js, SQLite

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Software Development, Database Systems, Distributed Systems, Data Mining, Machine Learning, Linear Algebra, Probability and Statistics, Statistical Modeling, Optimization

PROFESSIONAL EXPERIENCE

Stanford University

Software Engineering & Research Intern (School of Engineering)

Stanford, CA

Jan 2026 – April 2026

- Built and optimized **computational systems** that accelerated microfluidic experiment analysis by **50%**, enabling parallel processing of **thousands of molecular interactions**
- Implemented machine learning algorithms to estimate **kinetic and thermodynamic parameters** from experimental data, improving parameter accuracy by **30%** for protein interaction modeling
- Built automated **data acquisition tools** that increased experimental throughput and reduced manual errors in large-scale assays

University of Pennsylvania Perelman School of Medicine

Software Engineering & Medical Research Intern

Philadelphia, PA

May 2025 – August 2025

- Developed a **scalable Python-based MRI Spectroscopy analysis tool** replacing legacy **MATLAB-based** workflows, improving research efficiency by **40%**
- Designed **imaging pipelines** that processed **large-scale unstructured medical data**, reducing analysis time from hours to minutes for large-scale datasets
- Applied **statistical modeling** and **machine learning** to identify key patterns in brain imaging data, supporting **3 clinical research studies**

ProWeb Ltd

Software Engineering Intern

Remote

August 2024 – Dec 2024

- Resolved system-level issues **across cloud and local environments** for **200+ users**, reducing downtime by **65%**
- Developed automated SQL reporting pipelines that tracked **10 KPIs** and reduced manual reporting time by **10 hours** per week
- Developed internal tools that automated **4 manual support processes**, saving **12 hours** per week and improving workflow efficiency

Math & Computer Science Department - Stetson University

Research Assistant

DeLand, FL

September 2025 – Present

- Collaborated with **5 faculty members** on **Raspberry Pi projects**, developing **3 algorithms** for data preprocessing and experimental analysis using **Python and R**
- Developed **Python scripts** to automate data preprocessing and analysis workflows, reducing manual processing time by **40%**
- Mentored **50+** students in mathematics, leading to improvement in assignment completion rates and course performance

PERSONAL PROJECTS

Real-Time Environmental Streaming Pipeline | Python, Kafka, Spark, Airflow, PostgreSQL

- Developed a real-time streaming pipeline that ingests and processes **1,000+ sensor** data points per second using Kafka and Spark Structured Streaming
- Built Airflow-orchestrated ETL pipelines processing **1 million** data points daily into PostgreSQL, handling traffic spikes and enabling real-time analytics with **5-second** data freshness

MRI Frequency Shift Tool (UPenn-FIRE) | Python, FFT, Docker, Siemens FIRE

- Developed a Python-based MRI spectroscopy preprocessing tool using FFT, improving signal processing efficiency by **80%**
- Containerized the pipeline with **Docker**, reducing setup time across multiple environments

YouthLink Web Platform | Python, Flask, SQLite, HTML, CSS, Bootstrap

- Developed a **full-stack web platform** using **Flask and SQLite** to connect users with internships and opportunities
- Built database-backed search and responsive UI to improve accessibility across devices

EXTRACURRICULAR ACTIVITIES

Resident Assistant, Stetson University: Mentored **20+ students**; organized community event

February 2025 – Present